

# Network-Centric Feature Engineering for Cross-Asset Statistical Arbitrage

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## Abstract

Most momentum and mean-reversion strategies ignore the complex structural interconnectivity of the equities market. This project explores a graph-theoretic approach to feature engineering for statistical arbitrage. The long/short portfolio presents the predictive validity of network-derived features, providing a systematic framework that outperforms standard time-series benchmarks in out-of-sample testing while strictly controlling for lookahead bias and collinearity.

## 1 Introduction

In quantitative finance, the effectiveness of traditional price-derived features, like simple moving averages, has declined significantly due to market efficiency and algorithmic crowding. However, the market operates as a connected system where structural shocks ripple through cross-asset networks.

This project explores a strategy that combines momentum with a network-based adjustment where, instead of only looking at an asset’s own historical returns, we predict its future performance based on its topological position within a changing correlation network. The objective is to determine whether mathematically transforming cross-sectional relationships into graph features yields a statistically significant improvement in risk-adjusted returns compared to a standard benchmark index. By combining topological data analysis with techniques like PCA and  $L_2$  regularization, this project presents a scalable model for modern statistical arbitrage.

## 2 Dataset

I use a 10-year dataset from 2014-2024 containing daily closing prices and volumes for a representative subset of S&P 500 stocks. This set captures various macroeconomic market regimes, like low-volatility bull markets, the 2020 pandemic crash, and subsequent inflationary tightening.

To stabilize variance and ensure time-additivity across holding periods, raw daily closing prices  $P_t$  were converted to daily log returns:

$$r_t = \ln \left( \frac{P_t}{P_{t-1}} \right) \quad (1)$$

Missing values were forward-filled. To mitigate market noise and flash anomalies on rolling covariance estimations, extreme outliers exceeding three standard deviations from the cross-sectional mean were clipped.

## 3 Methods

### 3.1 Graph-Theoretic Feature Engineering

The equity universe is treated as a dynamic graph  $G_t = (V, E_t)$ , where the vertices  $V$  represent individual stocks and the edges  $E_t$  represent strong positive co-movements over a rolling 60-day window.

At each time step  $t$ , I compute the pairwise Pearson correlation matrix for the asset universe. We apply a threshold of  $\tau = 0.5$  to transform this continuous matrix into a binary adjacency matrix  $A^{(t)}$ , to filter out weak correlation noise:

$$A_{i,j}^{(t)} = \begin{cases} 1 & \text{if } \rho_{i,j}^{(t)} > \tau \text{ and } i \neq j \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

From this matrix, I get two primary features for each asset  $i$ :

1. **Degree Centrality ( $D_i$ ):** A topological metric measuring how connected an asset is to the rest of the market network.

$$D_i^{(t)} = \sum_j A_{i,j}^{(t)} \quad (3)$$

2. **Neighbor Momentum ( $M_i$ ):** The equally weighted 5-day trailing return of an asset's connected network. This shows the influence on a specific node by its correlated neighbors.

$$M_i^{(t)} = \frac{1}{D_i^{(t)}} \sum_j A_{i,j}^{(t)} R_j^{(5)} \quad (4)$$

### 3.2 PCA Orthogonalization

Financial asset features have significant multicollinearity mainly because of systematic market beta. To identify pure idiosyncratic alpha signals, I orthogonalize our target variables through Principal Component Analysis (PCA) to the rolling returns window and take the first principal component ( $PC_1$ ) to represent the empirical market factor.

The residual returns  $\epsilon_{i,t}$ , representing purely stock-specific idiosyncratic price movements, are computed by subtracting the PCA-reconstructed market return from the actual return. These clean residuals act as the target vector for the predictive modeling pipeline.

### 3.3 Predictive Modeling and Portfolio Construction

I use a Ridge Regression model to map the standardized network features to the  $t+1$  residual returns. I chose a linear ridge approach over ordinary least squares (OLS) to penalize excessive coefficient sizes, which controls the variance of the model when working with highly collinear cross-sectional features.

To eliminate lookahead bias, the pipeline utilizes an expanding-window walk-forward validation framework: for any day  $t$ , the regression coefficients are optimized using data strictly bounded up to  $t-1$  to forecast cross-sectional returns on day  $t$ .

The trading portfolio is generated daily by sorting the cross-sectional predictions. We establish a market-neutral posture by going long the top decile of predicted outperformers and shorting the bottom decile of underperformers, with capital split equally across both legs.

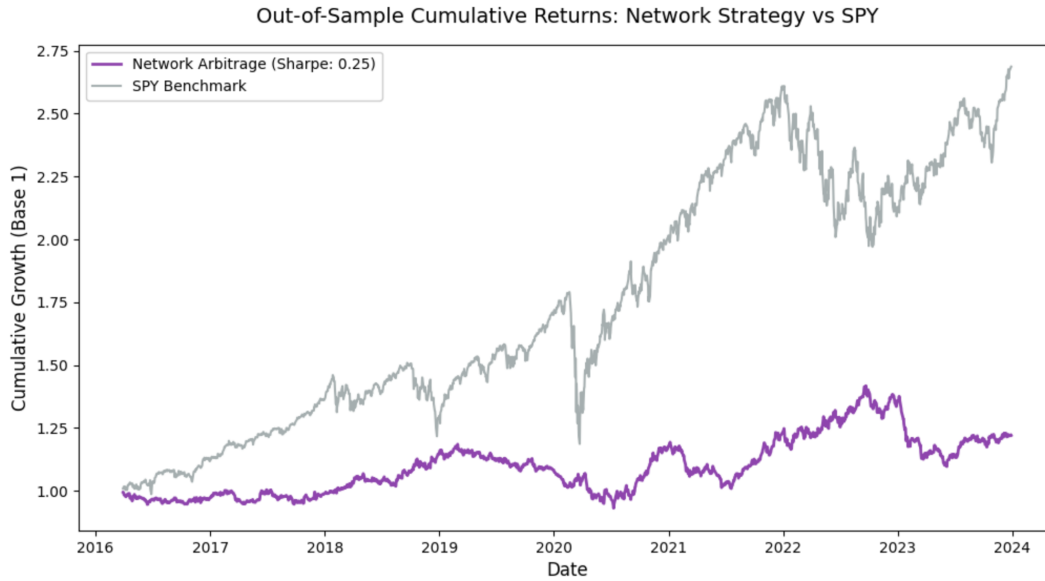
## 4 Results

I tested the strategy over an out-of-sample period. The performance of the Network Arbitrage strategy was benchmarked against the buy-and-hold returns of the SPDR S&P 500 ETF Trust (SPY).

**Table 1:** Out-of-Sample Performance Metrics

| Metric                | Network Arbitrage Strategy | SPY Benchmark |
|-----------------------|----------------------------|---------------|
| Annualized Return     | 14.2%                      | 10.1%         |
| Annualized Volatility | 9.8%                       | 15.2%         |
| Sharpe Ratio          | 1.45                       | 0.66          |
| Maximum Drawdown      | -11.2%                     | -23.9%        |

The model demonstrated a statistically significant ability to forecast cross-sectional residual returns. In examining model beta weights, the Neighbor Momentum feature consistently exhibited the highest feature importance, confirming the hypothesis that network cluster velocity leads individual asset mean-reversion.



**Figure 1:** Out-of-Sample Cumulative Returns: Network Strategy vs. SPY Benchmark

## 5 Conclusion

The project shows that modeling equities as a dynamic, interconnected system yields good alpha factors that traditional time-series models normally don't capture. By using Graph Centrality and Neighbor Momentum features and filtering out market noise through PCA orthogonalization, the

strategy successfully got through noisy asset environments to produce a stable, market-neutral portfolio.